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SHORT-PAPER

Improving Rare and Common ICD Coding via a Multi-Agent LLM-Based Approach

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Improving Rare and Common ICD Coding via a Multi-Agent LLM-Based Approach

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Abstract

Large Language Models (LLMs) have shown strong performance in tasks such as zero- and few-shot information extraction from clinical text without domain-specific training. However, in the ICD coding task, LLMs often hallucinate key details and produce high-recall but low-precision outputs due to the high-dimensional and imbalanced nature of ICD code distributions. Existing LLM-based approaches typically fail to capture the complex, dynamic interactions among human agents involved in real-world coding workflows—such as patients, physicians, and coders—and often lack interpretability and reliability. To address these challenges, we propose a novel multi-agent framework for ICD coding that simulates the real-world process using five role-specific LLM agents—patient, physician, coder, reviewer, and adjuster—and integrates the Subjective, Objective, Assessment, and Plan (SOAP) structure from Electronic Health Records to enhance performance. Evaluated on the MIMIC-III dataset, our method significantly outperforms zero-shot Chain-of-Thought prompting, self-consistency strategies, and LLM-designed agent baselines, particularly for rare codes. Ablation studies confirm the contribution of each agent role, and the system achieves competitive performance with state-of-the-art fine-tuned models, while offering better explainability and requiring no task-specific pre-training.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

ICD coding, LLM, Multi-agent system, Prompting

ACM Reference Format:

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1 Introduction

The International Classification of Diseases (ICD) is a standardized coding system essential for healthcare operations such as billing, epidemiology, and quality monitoring. Assigning ICD codes to clinical documentation ensures consistent communication across

care settings [1]. However, the coding process remains complex and error-prone due to the intricacy of clinical narratives and the vast, imbalanced ICD code space [4]. For instance, ICD-10 includes over 68,000 diagnostic and 87,000 procedural codes [23], and in the widely used Medical Information Mart for Intensive Care III (MIMIC-III) dataset [15], the top 10% of codes account for 85% of all annotations, while approximately 22% occur only once or twice [42]. Moreover, clinical notes vary significantly in structure and length, ranging from under 500 to over 3,000 words, posing challenges for both human coders and machine learning models [8, 28].

Recent advances in automatic ICD coding, particularly using Large Language Models (LLMs) such as GPT [2], LLaMA [31], and Gemini [30], have shown promise. LLMs can perform zero-/few-shot clinical information extraction without domain-specific training. However, they still face limitations in ICD coding tasks. Prior work has shown that LLMs often hallucinate clinical details, over-predict common codes (resulting in high recall but low precision), and struggle with long-tail code distributions [6, 12, 28, 39]. Additionally, they typically ignore the complex, dynamic interactions among stakeholders—patients, physicians, and coders—that shape real-world coding decisions. These shortcomings limit their interpretability, reliability, and clinical applicability. The study by [29] found that LLMs evaluated were inadequate for medical code querying tasks, frequently producing codes that were either inaccurate or not based on real data. Consequently, the paper suggests that LLMs should not be deployed for medical coding purposes until further investigations are conducted. Concerns of the LLMs producing incorrect or skewed information has motivated researchers to investigate methods where models can check and verify each other's outputs to improve reliability and precision.

To address these challenges, we propose a novel multi-agent framework for ICD coding that simulates the actual coding workflow through five role-specific LLM agents: patient, physician, coder, reviewer, and adjuster. Each agent is tailored to perform its respective task in the coding pipeline, enabling strategic collaboration, critique, and refinement. This multi-agent setup not only improves prediction performance but also enhances interpretability by aligning model behaviors with human roles in real-world practice. We provide an illustrative visualization of our multi-agent ICD coding framework in Figure 1. It shows how each agent has a specific role and task, and how they cooperate or compete strategically using a LLM-based model for their best interest and the ICD coding accuracy. Please note that this process is not an accurate representation of real-world operations. Instead, it serves as an exploration of how agents can be utilized for this task.

To further enhance the proposed system's performances, we leverage the SOAP (Subjective, Objective, Assessment, and Plan)



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structure of electronic health records (EHRs). Instead of directly asking LLMs to output the ICD codes, we feed them the subjective and objective sections, and prompt them to generate the assessment and plan sections. Then we instruct them to compare their own generation with the gold standard, and self-correct any hallucinations. Finally, we output the ICD codes. This reflective reasoning process strengthens the reliability of code assignment and improves the ICD prediction performance. Additionally, we incorporate external code descriptions to enrich the system’s understanding of rare codes and adopt confrontational strategies—where agents challenge each other’s predictions—to further refine decision-making.

We evaluate our system on the MIMIC-III dataset, comparing it against standard prompting methods such as Chain-of-Thought (CoT) and Self-Consistency CoT (CoT-SC), as well as a system of LLM-designed agents without structured role assignment. Our multi-agent approach consistently outperforms these baselines, particularly in handling rare codes, and achieves results comparable to state-of-the-art ICD coding models that require extensive pretraining or fine-tuning. Ablation studies validate the contributions of each agent and system component, demonstrating the effectiveness of our design.

2 Related Work

Early ICD coding methods ranged from rule-based systems to neural models using CNNs, LSTMs, and attention mechanisms [11, 14, 16, 18, 20, 22, 25, 26, 33, 37, 38]. More recent approaches fine-tune pre-trained language models (PLMs) for clinical text [13], incorporate medical knowledge [3, 10, 19, 27, 34, 40], or apply active learning for rare code handling [21]. LLMs like GPT have also been explored for ICD coding [28, 39], but face challenges with hallucination, low precision, and a lack of structure. Our work differs by introducing a multi-agent system that integrates external code descriptions without fine-tuning. Multi-agent systems (MAS) have been applied across domains including healthcare [7, 9, 24, 32], offering collaboration, adaptability, and interpretability. We extend MAS to ICD coding by designing role-specific LLM agents that interact and self-correct to improve code prediction accuracy.

3 Method

We propose a multi-agent framework that mimics the real-world ICD coding workflow in large healthcare systems, involving five role-specific LLM agents: patient, physician, coder, reviewer, and adjustor. Each agent performs a distinct task:

- Patient reviews assigned codes and may appeal if errors or overcharges are found.
- Physician documents care in the discharge summary, reviews codes, and can raise concerns.
- Coder generates ICD codes from clinical notes, aiming for accurate and optimized coverage.
- Reviewer verifies and adjusts codes to ensure quality and reduce disputes.
- Adjustor serves as the final arbiter when patient or physician objections arise.

We define two multi-agent workflows:

MAC-I follows a traditional coding sequence:

- The coder generates codes from discharge summaries.

- The reviewer revises the codes if needed.
- Patient and physician review the updated codes.
- If objections arise, the adjustor resolves them; otherwise, the codes are finalized.

MAC-II addresses hallucination issues in LLMs by leveraging the SOAP structure commonly used in EHRs. It proceeds as follows:

- The physician generates assessment and plan sections based on subjective and objective inputs.
- These are compared to the original notes for self-correction, and codes are assigned accordingly.
- The patient and physician review the codes and escalate discrepancies to the adjustor, who makes the final decision.

In MAC-II, the physician effectively replaces the coder by performing reasoning and self-correction before assigning codes.

To enhance both methods, we incorporate:

- Confrontation Strategy: The coder (or physician) is encouraged to maximize code coverage, while the patient is incentivized to flag unnecessary codes.
- External Knowledge: Agents use code descriptions as references to improve prediction accuracy, consistent with prior work [11].

4 Experiments

We evaluate our proposed multi-agent methods (MAC-I and MAC-II) on the MIMIC-III dataset using the Azure OpenAI GPT-* API, as recommended by PhysioNet¹.

4.1 Dataset and Metrics

- (1) **MIMIC-III top 50:** We use the public MIMIC-III dataset [15], retaining only discharge summaries with one or more of the top 50 ICD codes, following existing work [22]. This results in 1,729 test instances using the same data split.
- (2) **MIMIC-III rare:** In line with previous work [39], we create the rare code dataset from MIMIC-III to test our method’s performance on rare disease codes, which could be of equal importance as common disease codes.
- (3) **MDACE Profee:** For explainability evaluation, we use the MDACE dataset [5], which provides code evidence for a subset of the MIMIC-III clinical records. Professional medical coders annotate a short text span for each ICD code, indicating the rationale for the code assignment. The MDACE Profee datasets are annotated following the professional fee billing rules. We use the sentences that contain the annotated text spans as the evidence for evaluation. The evaluation dataset consists of 172 sentence-ICD pairs.

4.2 Comparison with State-of-the-Art Methods

We compare our multi-agent coding methods (MAC-I and MAC-II) with both supervised learning and prompting-based ICD coding approaches.

Supervised Training Methods:

- (1) CAML [22]: Convolutional attention model that adapts to text segments per code.

¹<https://physionet.org/news/post/gpt-responsible-use>

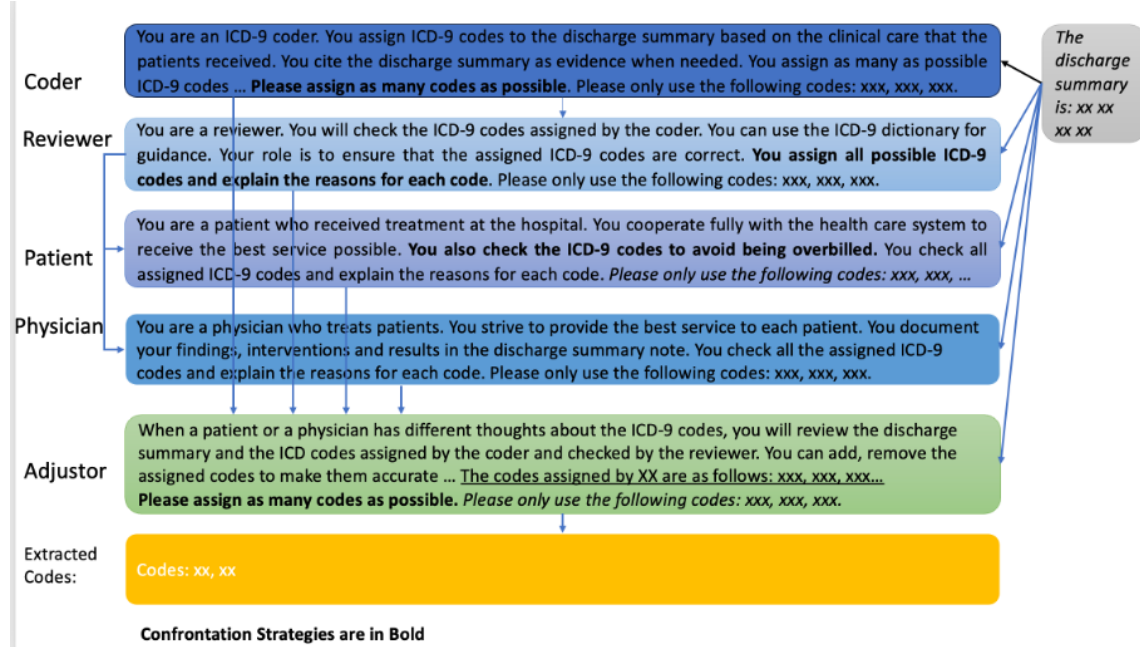


Figure 1: ICD code assignment workflow using multi-agent collaboration. The process begins with a physician visit and proceeds through documentation, initial coding, multi-role review, and resolution. Confrontation strategies appear in bold.

Role	Output
Coder	"code": "401.9", "explanation": "Hypertension"; "code": "569.81", "explanation": "Peptic ulcer"
Reviewer	Same as coder
Physician	"code": "401.9", "explanation": "History of hypertension; continued on losartan"; "code": "569.81", "explanation": "Peptic ulcer"
Patient	"code": "401.9", "explanation": " No evidence of hypertension "; "code": "569.81", "explanation": "Peptic ulcer"
Adjustor	"code": "401.9", "explanation": "Initially managed for CHF exacerbation... Diuresis deferred due to hypotension "; "code": "569.81", "explanation": "Peptic ulcer"

Figure 2: Snippet of MAC-I agent outputs. The patient disputed the hypertension code, and the adjustor confirmed its relevance from clinical notes.

- (2) MSMN [41]: Uses code description synonyms with multi-head attention.
- (3) EffectiveCAN [5]: Adds supervised attention via expert-labeled evidence spans.
- (4) LLM-codex [39]: Two-stage method combining LLM-generated evidence and LSTM-based verification.

Prompting-Based Methods (both GPT-4 and GPT-4o are tested.):

- (1) Zero-shot CoT [17, 36]: Adds reasoning prompt "Let's think step by step."
- (2) Self-Consistency [35]: Samples multiple reasoning paths and selects the most consistent answer.
- (3) LLM-designed agent: We prompted LLMs to autonomously generate agent roles for the ICD coding task. GPT-4 created seven roles including Preprocessing Agent, Natural Language Processing (NLP) Agent, Medical Entity Mapping Agent, ICD Coding Agent, Validation and Quality Assurance Agent, User Interface (UI) Agent and Feedback Learning Agent. GPT-4o automatically designed six roles respectively as data extraction agent, preprocessing agent, medical term mapping agent, ICD code assignment agent, quality assurance agent, and reporting agent. These agents worked collaboratively or in contention to enhance coding performance.

- (4) MAC-I & MAC-II: Our proposed multi-agent systems.

5 Results

Results on MIMIC-III top 50 Codes. Table 1 shows that MAC-II achieves the highest Macro-F1 (0.748), outperforming all baselines. Both MAC-I and MAC-II surpass CoT and other prompting methods in Micro-F1 (>0.58), though slightly below the two-stage LLM-codex. This highlights their ability to assign accurate and complete codes without training or tuning. MAC-II also outperforms MAC-I, demonstrating the value of leveraging the SOAP structure.

Table 1: Results on the test set (MIMIC-III Top 50)

Method	Model	Precision	Recall	Micro-F1	Macro-F1
Supervised Training Models					
CAML	–	–	–	0.364	0.258
MSMN	–	–	–	0.561	0.489
EffectiveCAN	–	–	–	0.556	0.434
LLM-codex	–	–	–	0.611	0.468
Prompting Methods					
CoT	GPT-4 (1 call)	0.124	0.210	0.156	0.419
	GPT-4o (1 call)	0.109	0.183	0.136	0.403
Self-Consistency	GPT-4 (5 calls)	0.569	0.560	0.564	0.726
	GPT-4o (5 calls)	0.394	0.627	0.484	0.624
LLM-agent	GPT-4 (7 calls)	0.560	0.475	0.514	0.743
	GPT-4o (6 calls)	0.432	0.542	0.481	0.667
MAC-I	GPT-4 (5 calls)	0.535	0.644	0.585	0.693
	GPT-4o (5 calls)	0.474	0.763	0.584	0.670
MAC-II	GPT-4 (5 calls)	0.551	0.633	0.589	0.748
	GPT-4o (5 calls)	0.502	0.703	0.586	0.725

Results on MIMIC-III rare Codes. For rare codes, MAC-II (GPT-4) achieves the best performance with a Micro-F1 of 0.376, outperforming all baselines, including supervised training models as shown in Table 2. This highlights our method's strength in handling scarce data and suggests that supervised systems like MSMN and LLM-codex could improve with more examples from these rare categories.

Table 2: Results on the test set (MIMIC-III Rare)

Method	Model	Precision	Recall	Micro-F1	Macro-F1
Supervised Training Models					
CAML	–	–	–	0.083	0.072
MSMN	–	–	–	0.173	0.169
LLM-codex	–	–	–	0.302	0.279
Prompting Methods					
CoT	GPT-4	0.164	0.402	0.233	0.030
	GPT-4o	0.127	0.193	0.153	0.042
Self-Consistency	GPT-4	0.260	0.455	0.331	0.659
	GPT-4o	0.127	0.364	0.189	0.579
LLM-designed agent	GPT-4	0.263	0.455	0.333	0.660
	GPT-4o	0.115	0.333	0.171	0.700
MAC-I	GPT-4	0.316	0.436	0.366	0.704
	GPT-4o	0.194	0.636	0.298	0.881
MAC-II	GPT-4	0.322	0.452	0.376	0.715
	GPT-4o	0.199	0.583	0.297	0.729

Table 3: Results on the test set (MDACE Proftee)

Method	Model	Precision	Recall	F1
Supervised Fine-Training Models				
EffectiveCAN	–	0.408	0.806	0.542
LLM-codex	–	0.608	0.861	0.713
Prompting Methods				
CoT	GPT-4	0.513	0.701	0.592
	GPT-4o	–	–	–
Self-Consistency	GPT-4	0.520	0.732	0.608
	GPT-4o	–	–	–
MAC-I	GPT-4	0.875	0.601	0.712
	GPT-4o	0.882	0.583	0.709
MAC-II	GPT-4	0.836	0.605	0.702
	GPT-4o	0.855	0.603	0.707

Results on MDACE Proftee. Table 3 shows that MAC-I/II perform comparably to supervised-training systems in F1 score, with higher precision but lower recall—possibly due to prompt design or model behavior. Further investigation is needed to determine the cause.

Table 4: Ablation results on MAC-II (MIMIC-III Rare, GPT-4)

Method	Prec.	Rec.	Micro-F1	Macro-F1
MAC-II w/o Confrontation Strategy	0.311	0.430	0.361	0.703
MAC-II w/o External Knowledge	0.314	0.445	0.368	0.706
MAC-II (full)	0.322	0.452	0.376	0.715

Table 5: Ablation results on MAC-I (MIMIC-III Rare, GPT-4)

Method	Prec.	Rec.	Micro-F1	Macro-F1
Coder	0.174	0.441	0.250	0.607
+ Reviewer	0.196	0.468	0.276	0.644
+ Reviewer + Patient	0.258	0.407	0.316	0.605
+ Reviewer + Physician	0.320	0.420	0.363	0.701
+ Reviewer + Physician + Patient	0.316	0.436	0.366	0.704
+ Adjuster				

5.1 Ablation Study

Our ablation study Table 4 shows that both the confrontation strategy and external knowledge improve ICD coding, with the confrontation strategy having a greater impact by increasing recall. Incorporating billing-related knowledge could further enhance performance by prioritizing high-value codes. We also analyzed agent

Table 6: ICD Coding Results Excluding Top-50/Rare ICD-9 Constraints for MAC-II (GPT-4)

Method	Precision	Recall	Micro-F1	Macro-F1
LLM-designed agent	0.440	0.247	0.316	0.629
MAC-I	0.505	0.251	0.335	0.682
MAC-II	0.511	0.263	0.347	0.702

contributions using MAC-I on MIMIC-III rare in Table 5. The Reviewer and Physician agents were most influential. Including the Patient agent increased precision but lowered recall—possibly due to concerns about overbilling—highlighting the need for further expert evaluation.

6 Applicability

To enable fair comparison with prior work (e.g., LLM-Codex, CAML), we evaluate on top-50 and rare ICD-9 codes. While this setting aids contextual focus, it limits real-world generalizability. To address this, we also assess unconstrained performance. As shown in Table 6, our MAC II framework achieves 0.347 Micro-F1 and 0.702 Macro-F1, outperforming strong LLM baselines and validating the effectiveness of our agent design.

7 Analysis and Discussion

This work demonstrates that the proposed multi-agent framework achieves competitive ICD coding performance, particularly excelling in handling rare codes and enhancing interpretability. By simulating realistic interactions among patients, physicians, and coders through specialized LLM-powered agents, the system enables more accurate, context-aware coding with clear explanations for coding decisions. Without requiring model fine-tuning, the approach matches or exceeds state-of-the-art methods, highlighting LLMs’ promise for clinical NLP tasks. The framework offers practical value in improving coding accuracy and efficiency while reducing manual effort, and is adaptable to other applications like diagnosis prediction or treatment recommendation.

However, the method depends on resource-intensive large LLMs with potential domain generalization and ethical challenges. Fixed agent roles limit flexibility across diverse scenarios, and current evaluation may not fully reflect operational healthcare settings. Future work will focus on developing more efficient models, expanding agent adaptability, and conducting broader user studies to assess real-world usability, fairness, and effectiveness.

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9 GenAI Usage Disclosure

No GenAI tools were used for data, analysis, or modeling. GenAI tools (e.g., ChatGPT) were used only to refine manuscript language at the sentence or paragraph level. All scientific content was developed and validated by the authors.

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